

Research Notes on Stable Diffusion

Stable Diffusion

$$\mathcal{L} = \mathbb{E}_{x, \epsilon, t} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]$$

>> Section 1

Learning Transferable Visual Models From Natural Language Supervision (2021). This paper describes the CLIP method for training text encoders, which convert text into floating point vectors. Such text encodings are used by the diffusion model to create images. SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations (2021). This paper describes SDEdit, aka "img2img". High-Resolution Image Synthesis with Latent Diffusion Models (2021, updated in 2022). This paper describes the latent diffusion model (LDM). This is the backbone of the Stable Diffusion architecture. Classifier-Free Diffusion Guidance (2022). This paper describes CFG, which allows the text encoding vector to steer the diffusion model towards creating the image described by the text. SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis (2023). Describes SDXL. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow (2022). Describes rectified flow, which is used for the backbone architecture of SD 3.0. Scaling Rectified Flow Transformers for High-resolution Image Synthesis (2024). Describes SD 3.0. Training cost

>> Section 2

The text to image sampling script within Stable Diffusion, known as "txt2img", consumes a text prompt in addition to assorted option parameters covering sampling types, output image dimensions, and seed values. The script outputs an image file based on the model's interpretation of the prompt. Generated images are tagged with an invisible digital watermark to allow users to identify an image as generated by Stable Diffusion, although this watermark loses its efficacy if the image is resized or rotated. Each txt2img generation will involve a specific seed value which affects the output image. Users may opt to randomize the seed in order to explore different generated outputs, or use the same seed to obtain the same image output as a previously generated image. Users are also able to adjust the number of inference steps for the sampler; a higher value takes a longer duration of time, however a smaller value may result in visual defects. Another configurable option, the classifier-free guidance scale value, allows the user to adjust how closely the output image adheres to the prompt. More experimental use cases may opt for a lower scale value, while use cases aiming for more specific outputs may use a higher value.

Additional text2img features are provided by front-end implementations of Stable Diffusion, which allow users to modify the weight given to specific parts of the text prompt. Emphasis markers allow users to add or reduce emphasis to keywords by enclosing them with brackets. An alternative method of adjusting weight to parts of the prompt are "negative prompts". Negative prompts are a feature included in some front-end implementations, including Stability AI's own DreamStudio cloud service, and allow the user to specify prompts which the model should avoid during image generation. The specified prompts may be undesirable image features that would otherwise be present within image outputs due to the positive prompts provided by the user, or due to how the model was originally trained, with mangled human hands being a common example.

>> Section 3

Andersen, McKernan, and Ortiz v. Stability AI, Midjourney, and DeviantArt In January 2023, three artists, Sarah Andersen, Kelly McKernan, and Karla Ortiz, filed a copyright infringement lawsuit against Stability AI, Midjourney, and DeviantArt, claiming that these companies have infringed the rights of millions of artists by training AI tools on five billion images scraped from the web without the consent of the original artists. In July 2023, U.S. District Judge William Orrick inclined to dismiss most of the lawsuit filed by Andersen, McKernan, and Ortiz but allowed them to file a new complaint, providing them an opportunity to reframe their arguments.

>> Section 4

Getty Images asserted that the training and development of Stable Diffusion involved the unauthorized use of its images, which were downloaded on servers and computers that were potentially in the UK. However, Stability AI argued that all training and development took place outside the UK, specifically in U.S. data centers operated by Amazon Web Services. Stability AI applied for reverse summary judgment and/or strike out of two claims: the training and development claim, and the secondary infringement of copyright claim. The High Court, however, refused to strike out these claims, allowing them to proceed to trial. The court is to determine whether the training and development of Stable Diffusion occurred in the UK, which is crucial for establishing jurisdiction under the UK's Copyright, Designs and Patents Act 1988 (CDPA). The secondary infringement claim revolved around whether the pre-trained Stable Diffusion software, made available in the UK through platforms like GitHub, HuggingFace, and DreamStudio, constitutes an "article" under sections 22 and 23 of the CDPA. As of November 4, 2025 Getty Images largely lost the lawsuit over AI image

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generator to Stability AI, prompting Getty and some lawyers to call for stronger protections for copyright owners in Britain.

>> Section 5

Capabilities The Stable Diffusion model supports the ability to generate new images from scratch through the use of a text prompt describing elements to be included or omitted from the output. Existing images can be re-drawn by the model to incorporate new elements described by a text prompt (a process known as "guided image synthesis") through its diffusion-denoising mechanism. In addition, the model also allows the use of prompts to partially alter existing images via inpainting and outpainting, when used with an appropriate user interface that supports such features, of which numerous different open source implementations exist. Stable Diffusion is recommended to be run with 10 GB or more VRAM, however users with less VRAM may opt to load the weights in float16 precision instead of the default float32 to tradeoff model performance with lower VRAM usage.

>> Section 6

Training data Stable Diffusion was trained on pairs of images and captions taken from LAION-5B, a publicly available dataset derived from Common Crawl data scraped from the web, where 5 billion image-text pairs were classified based on language and filtered into separate datasets by resolution, a predicted likelihood of containing a watermark, and predicted "aesthetic" score (e.g. subjective visual quality). The dataset was created by LAION, a German non-profit which receives funding from Stability AI. The Stable Diffusion model was trained on three subsets of LAION-5B: laion2B-en, laion-high-resolution, and laion-aesthetics v2 5+. A third-party analysis of the model's training data identified that out of a smaller subset of 12 million images taken from the original wider dataset used, approximately 47% of the sample size of images came from 100 different domains, with Pinterest taking up 8.5% of the subset, followed by websites such as WordPress, Blogspot, Flickr, DeviantArt and Wikimedia Commons. An investigation by Bayerischer Rundfunk showed that LAION's datasets, hosted on Hugging Face, contain large amounts of private and sensitive data.